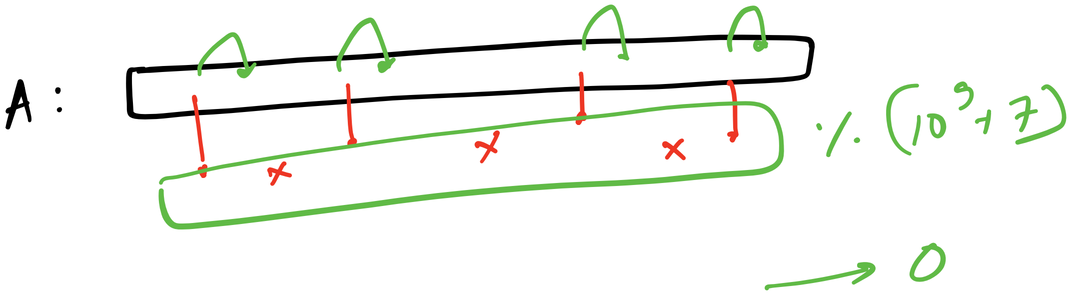
Q1. No. Game & friend-2



$A : [7, 7, 4, 5, 3]$ (× ✓ × ✓ ×)

$(35) \% \ 10^9+7$

↓

35

$A[i] <= 10^9$
↓
int

```
int M = 10^9+7;
~~int~~ long p = 1;     bool ok = false;
f( i=0; i<N-1; i++){
    if( A[i] > A[i+1]){
        ok = true;
        p = ((p ~~% M~~) × (A[i] ~~% M~~)) % M;
    }
}
if( ok == false)
    p = 0;
ret p;
```

$p = (p \times A[i]) \% M;$

$\cancel{i} \to l$, i

$\sim 10^9 \times \sim 10^9 \to \sim 10^{18}$

Q Alternating Bits

I

A : 5 → 101 → ✓

A : 10 → 1010 → ✓

A : 20 → 10100 → ✗

A : 7 → 111 → ✗

A : 32 0000100000 → ✗

✗

```
      3 2 1 0
A : 10 → 1 0 1 0        (A&1)
       & 0 0 0 1
       ---------
               0   →   0
```

A : A>>1

```
   0 1 0 1
 & 0 0 0 1
 ---------
         1   →   1     (A&1)
```

A = A >> 1

(A & 1)

$$
\begin{array}{rcccc}
 & 0 & 0 & 1 & 0 \\
\& & 0 & 0 & 0 & 1 \\
\hline
 & & & & 0
\end{array}
\rightarrow 0
$$

A = A >> 1

(A & 1)

$$
\begin{array}{rcccc}
 & 0 & 0 & 0 & 1 \\
\& & 0 & 0 & 0 & 1 \\
\hline
 & & & & 1
\end{array}
\rightarrow 1
$$

A = A >> 1

$\boxed{0\ 0\ 0\ 0} \rightarrow$ STOP

// A

```
while (A > 0) {
    int curBit = (A & 1);

    A = A >> 1;

}
```

rt f

rl f

II

```
            31              0
                3   2   1   0
A : 10  →       1   0   1   0
                mx
```

```
mx = 0;
f(i = 31; i >= 0; i--) {
    if ((A & (1<<i)) > 0) {
        mx = i;
        break;
```

}

}

mx

mx-1

-2

3 2 1 0

0 ---- 0 0 1 0 1 0

-2

```
f(i = mx; i >= 0; i -= 2){
    if((A & (1<<i)) == 0){
        ret false;
    }
}
f(i = mx-1; i >= 0; i -= 2){
    if((A & (1<<i)) > 0){
        ret false;
    }
}
ret true;
```

Q3 Tens Digit Sorting

A

→ INC order of tens place — does not exist : 0

→ tie : larger num before smaller num.

02

A: [2, 24, 22, 19]

tens: 0 2 2 1

Sorted

A: [2, 19, 24, 22]

int x = 1234; $\xrightarrow{x \% 10}$ 4 → 1's place

x/10

123 $\xrightarrow{x \% 10}$ 3 → 10's place

```
int findTensDigit( int x){
    ret ((x/10)%10);

}

bool comp ( int a, int b) {
    int dA = findTensDigit(a);
    int dB = ____________(b);
    if ( (dA < dB) || ((dA == dB) && a>b)){
        ret true;

    }
    ret false;

}

A.sort(comp);
```